

Revision Report / Response to Reviewers

Initial Placement for Fruchterman–Reingold Force Model With Coordinate Newton Direction

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Stephen Kobourov
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Dear Prof. Stephen Kobourov,

Thank you for directing us to the reviewers’ comments and for giving us the opportunity to revise our manuscript. We confirm that we have received the review reports.

Based on the reports, we have carefully reviewed all comments and revised the manuscript accordingly. Our responses are provided below in a point-by-point manner. Where applicable, we refer to section, equation, figure, or algorithm numbers in the revised manuscript.

We hope the revised version is now suitable for publication and look forward to hearing from you.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda

A Response to Reviewer A

We thank Reviewer A for the careful and constructive assessment of our manuscript. We appreciate the reviewer’s observation that the proposed algorithm is interesting and that there are no major problems with the construction of the algorithm. We have revised the manuscript to address the concerns regarding the relationship between the proposed method and the Fruchterman–Reingold model, as well as the distinction between the FR and KK models.

A.1 Relationship Between the Proposed Approach and the FR Force Model

First, the authors do not sufficiently consider how well their approach matches the FR force model.

Originally, Fruchterman and Reingold [10] proposed the FR model and solved it by using a pseudo-physical simulation-based algorithm (by following Eades’ spring embedder method [8]). By contrast, Kamada and Kawai [21] proposed the KK force model and solved it by a Newton-like method. Also, many succeeding studies followed these approaches, i.e., either by combining simulation-based algorithms with the FR model or by combining Newton-like methods with the KK model. Although Hosobe [19] showed that both models could be solved with a Newton-like method called L-BFGS, the results still suggested that the KK model was more suitable for L-BFGS than the FR model.

However, this submitted paper adopts an unusual approach that combines a Newton-like method with the FR model. I understand that the paper is focused only on the

computation of initial layouts, but I think that the validation of adopting this approach is still insufficient.

Response: Thank you for the detailed comment. We do not disagree with these observations themselves.

It should be noted that the experimental results in Section 5 show that the proposed initial layout algorithm works better when it is combined with the L-BFGS algorithm for computing final layouts.

Response: First, we recognize that the previous version of the manuscript contained several expressions that could have caused misunderstanding, and we apologize for this.

As a premise, Reviewer A’s comment mainly concerns the comparison between algorithms used after initialization, namely the FR algorithm versus L-BFGS. However, what we truly intend to claim is a comparison of initialization methods: Random initialization versus Coordinate Newton initialization, or SA-based initialization versus Coordinate Newton initialization. Therefore, the observation that “the proposed initial layout algorithm works better when it is combined with the L-BFGS algorithm for computing final layouts” is attributable to the difference between the non-proposed components, and is not directly related to the superiority or inferiority of the proposed method itself.

We considered carefully how best to respond to this point, and revised the manuscript in two directions.

First, as stated above, we made it clear that the point raised concerns only the difference between existing post-initialization methods and is independent of the merit of the proposed method. It is true that the result labeled CN-LBFGS gives the best performance and is better than CN-FR. However, this reflects the difference between existing optimization methods, which is not the central focus of this paper. What we want readers to compare is the difference between RAND-FR and CN-FR, or between RAND-LBFGS and CN-LBFGS. We clarified in Section 5 that this is the comparison on which attention should be focused.

Second, we revised expressions that could have caused the misunderstanding above. In the previous manuscript, as Reviewer B also noted, we referred to the superiority of L-BFGS in the introduction and elsewhere, which may have given the impression that L-BFGS itself was part of the proposed method. However, as illustrated for example in Fig. 1, the proposed method and L-BFGS are completely independent components. If a reader wishes to use the FR algorithm, they may do so; indeed, using the proposed method in combination with FR is also within the intended scope of the method. We revised the wording in the introduction and related parts so that this distinction is communicated more clearly.

The discussion continues in the next response.

It seems from Figure 6 that the combination of the proposed initial layout algorithm with the FR algorithm for computing final layouts does not work well except in limited cases like jagmesh1.

Response: Related to the preceding comment, we believe that this particular point is based on a misunderstanding. In Figure 6, the proposed method outperforms the baseline in all cases except jagmesh1, and we believe that the differences are sufficiently significant both numerically and visually.

We suspect that the way Figure 6 was presented may have made it difficult to interpret correctly. To prevent such misunderstandings, we emphasized the proposed method in bold, explicitly added the label “(proposed)” to the CN method, and reordered the columns so that the comparison between random initialization and the proposed method is easier to see. We also clarified this point in the main text.

If the authors think that the proposed algorithm should be combined with the L-BFGS, they should adopt the KK model instead of the FR model, which would be more natural and more persuasive.

Response: As the explanation above indicates, our position is that the proposed method is effective whether it is combined with the FR algorithm or with L-BFGS. Therefore, we do not argue that it “should be” combined specifically with L-BFGS.

In fact, in many cases the difference between Rand-FR and CN-FR is larger than the difference between Rand-LBFGS and CN-LBFGS. In this sense, the effect of the proposed method is often more pronounced when it is combined with the FR algorithm. Therefore, at this stage we have not explicitly adopted the suggestion that the KK model should be used instead. As discussed later, however, we regard combinations with other force models and generalizations in that direction as valuable topics, and we added a separate subsection to discuss them.

At the same time, for readers who simply want to obtain the best final result, it is fair to say as a general observation based on prior work, independently of the proposed method, that L-BFGS often gives better results than the FR algorithm as an optimization method. We hope that clarifying this point adequately addresses the reviewer’s concern.

If the authors think that the proposed algorithm works well for the FR layout algorithm (or another simulation-based algorithm), they should more deeply explore this direction and present clearer results in the paper.

Response: To reiterate, our claim is not that the proposed method should be combined with the FR algorithm or should be combined with L-BFGS. Rather, both combinations are within the scope of the proposed algorithm. As stated above, we revised the manuscript to make this point clear.

A.2 Difference Between FR and KK Models

Second, the authors do not sufficiently consider differences between the FR force model and others, especially the KK model, which becomes apparent especially in Subsection 7.1.

Here the authors first say “Although we utilized the coordinate Newton direction only for the optimization problem (1), we can see that its application is not necessarily limited to the FR force model alone”, and then present “objective functions arising from graphs”. Does the authors mean that this discussion applies to other force models including the KK model? If so, I think that the authors should adopt the KK model instead of the FR model.

Response: Thank you for this insightful comment. Our response to this point has two parts.

First, as the reviewer suggests, the proposed method is applicable to other force models such as KK, HC, and Eades. This is because the coordinate Newton direction can be defined in the

same way for functions of this form. Second, however, for the KK model, the effect of the proposed method may be less pronounced than for other models, or it may even be counterproductive, because of the specific characteristics of the KK model.

We first discuss the generalization to the HC and Eades layouts. These force models are also mentioned in Hosobe [19], and the proposed method is indeed applicable to them. We also conducted numerical experiments. As described in detail in the revised manuscript, some additional care is required because of differences such as the presence or absence of convexity, but the proposed method was found to be effective in essentially the same way. We added explanations and numerical results on these points to the main text and the Appendix. Since these models are somewhat less widely known than the FR model, we refer to them only briefly in the introduction. We also considered changing the title of the paper, but decided to keep it unchanged.

In contrast, for the KK model, we believe that the effect of the proposed method may be less apparent than for other models. This is closely related to whether the sparsity of the graph can be exploited. The proposed method is designed to compute the force on each vertex by looking at the vertices adjacent to it and to optimize the layout based on this computation. Therefore, when the main terms in the energy function are at most $\mathcal{O}(|E|)$, that is, proportional to the number of edges, the proposed method can work much faster than naive methods requiring $\mathcal{O}(|V|^2)$ computations, and its effect is likely to be significant. This is a common feature of the FR, HC, and Eades models. Indeed, the examples of “objective functions arising from graphs” also have this form.

In the KK model, however, the main energy terms are defined for all pairs of vertices, rather than as a separate set of $\mathcal{O}(|E|)$ terms. This is because the energy in the KK model is defined by first computing all-pairs graph distances and then defining energy terms based on those distances. Therefore, computing forces on a vertex-by-vertex basis does not provide the same advantage and may even cause slower convergence.

As related work, a method based on another optimization approach called SGD has been proposed for the KK model: [33] J. X. Zheng, S. Pawar, and D. F. M. Goodman, “Graph Drawing by Stochastic Gradient Descent,” IEEE Transactions on Visualization and Computer Graphics. In contrast to the proposed method, that approach focuses on pairs rather than vertices when performing optimization, and in that sense it can be regarded as complementary to our work.

Based on these considerations, we removed the previous Section 7.1 and instead revised the manuscript, especially around Section 4.5, to explain these issues more carefully.

In addition, the authors present the graph isomorphism problem as an application of the proposed algorithm. I think that the KK model would be more suitable for this application than the FR model.

Response: Thank you for pointing this out. Regarding the application to graph isomorphism, we agree with the reviewer and recognize that our previous discussion reflected an insufficient understanding. This application is not particularly natural for the FR model and may distract readers from the main contribution of the paper. Therefore, we removed this discussion from the revised manuscript.

The following might work as hints for the authors’ paper revision. Intuitively, the FR model exhibits soft behavior while the KK model exhibits hard behavior. Therefore, the FR model is usually used for dynamic graph layouts like animation and interaction while the KK model is usually used for static layouts. Also, the FR model is sometimes used for very large graphs because simulation-based algorithms for the FR model can be more

scalable than Newton-like algorithms for the KK model. One plausible direction for the authors might be to combine the proposed algorithm with a scalable simulation-based algorithm for the FR model and to apply it to very large graphs.

Response: We sincerely thank the reviewer for providing this valuable hint. We believe that the soft behavior of the FR model is a very important perspective, especially in dynamic graph drawing scenarios such as animation and interaction. We strongly agree with this viewpoint, and such applications were among the motivations for this study. We added a discussion of this point in Section 6.1, including its relationship to the proposed initial placement method.

A.3 Notation

Symbol E is used for two different purposes, one to indicate a set of graph edges and the other to indicate energies between nodes possibly with subscripts and superscripts. It will be better to use different symbols for these different concepts.

Response: Thank you for pointing this out. We avoided the symbol conflict by using a dedicated symbol for the edge set, so that the energy terms $U_{i,j}$ are not confused with the edge set.

I wonder if $(i, j) \in V^2$ with $i \neq j$ in (1) is appropriately intended. This treats both (i, j) and (j, i) in the sum.

Response: We intended the sum to be over all ordered pairs of distinct vertices, so that both (i, j) and (j, i) are included. We clarified this in the text and notation.

Symbol ϵ appears only once in Subsection 4.4, which seems strange.

Response: We clarified the role of ϵ as the minimum separation in the discrete point set Q and its use in preventing divergence.

Symbol f_i at line 6 of Algorithm 1 probably lacks the superscript “a”.

Response: We corrected the missing superscript in Algorithm 1. The update now uses the attractive-only objective f_i^a .

In Subsection 5.3, the word “L-BFGS” that appears as superscripts of N does not seem to be correctly specified. The hyphen mark looks too long.

Response: Regarding the reviewer’s note on the “L-BFGS” superscripts: the hyphen was already intended as a proper hyphen in math mode, but its appearance could still be misleading. To avoid confusion, we now use the spelling “LBFGS” when it appears inside math expressions, for example in superscripts, while keeping the standard spelling “L-BFGS” in running text.

In the optimization problem called “objective functions arising from graphs” in Subsection 7.1, “=” probably should be “:=”.

Response: We corrected the definition symbol in the “objective functions arising from graphs” formulation.

The header of Subsection 8.1 seems to have a problem with the hyphen mark again.

Response: We checked the subsection headers and ensured that hyphenation uses proper LaTeX conventions.

B Response to Reviewer B

We thank Reviewer B for the careful review and constructive suggestions. We appreciate the positive assessment that the paper addresses a topic of clear interest to the JGAA audience, that the technical explanation is comprehensive, and that the mathematical treatment is mostly easy to follow. We have revised the manuscript to improve the framing of the contribution, the discussion of related work, the experimental evaluation, and the organization of the later sections.

B.1 Related Work on Multilevel Approaches

In the Related Work section, in 3.1 Multilevel approaches are quickly dismissed. I don't understand the statement the statement about sfdp being "prior work" to FR - as the latter came way later than the former. This sentence should be clarified.

Response: Thank you for pointing this out. We understand this comment as arising from a possible ambiguity in the term FR: whether it refers to the force model, which defines attractive and repulsive forces between vertices, or to an algorithm for finding an equilibrium point of that model. It is important to emphasize that, in the previous manuscript, the FR force model and the FR algorithm were treated as entirely different concepts. We apologize that the wording was highly prone to misunderstanding.

In response to this point, we revised the wording throughout the manuscript so that this distinction is conveyed clearly. We continue to refer to the force model itself, namely the formulation corresponding to Problem (1), as the FR force model, which is also the term used in the title. On the other hand, the algorithm used to find an equilibrium point of this model, previously called the FR algorithm in Algorithm 3, is now referred to as a simulation-based algorithm, and we explain clearly what this means. Since this algorithm is also commonly called the FR algorithm, we added an explanation while taking care to avoid confusion.

Thus, the statement that sfdp is prior work to FR was intended to refer only to the algorithmic aspect of sfdp, and not to the modeling aspect. As the reviewer points out, it was not intended to refer to the FR force model. We carefully revised the relevant passages and related parts of the manuscript.

To reiterate, the FR force model itself and the simulation-based algorithm used to solve it are entirely different objects.

B.2 Discussion of Multilevel Approaches and Potential Applications

Also, I believe that further space could be given to multi-level approaches, also considering that could be one potential application/extension of this technique beyond this paper.

Response: Thank you for the suggestion. As the reviewer notes, we believe that multilevel approaches are a highly promising direction for applying the proposed method. We added a new Section 6.1 to discuss this point.

B.3 Uncertain Framing of FR Versus L-BFGS

The framing of this paper is somewhat uncertain. Despite the paper title and general orientation towards FR, L-BFGS is consistently and repeatedly mentioned as a better alternative.

Response: We believe that this point also stems from the ambiguity mentioned above, namely the possible confusion between the FR algorithm and the FR model. We apologize for the wording that may have caused this misunderstanding. The title of the paper refers to the FR force model, namely the formulation in Problem (1), and not to the algorithm. Therefore, the relevant statements in the previous manuscript were intended to mean that L-BFGS often gives better results than the simulation-based algorithm, not that it is a better alternative to the FR force model itself. The target of this paper, including its title, is the optimization problem defined by the FR force model; the FR algorithm is only one possible method for solving that problem.

Moreover, whether the FR algorithm or L-BFGS is better is a difference between existing methods and is independent of the merit of the proposed method. In the previous manuscript, as Reviewer A also noted, our references to the superiority of L-BFGS in the introduction and elsewhere may have given the impression that L-BFGS itself was part of the proposed method. However, as shown for example in Fig. 1, the proposed method and L-BFGS are completely independent components. If a reader wishes to use the simulation-based algorithm, they may do so; indeed, combining the proposed method with the simulation-based algorithm is also within the intended scope of the method. We revised the introduction and related passages to make this distinction clearer.

Even in the related work section, in Section 3.1, only L-BFGS is mentioned as a ready-to-go measure for application in Multilevel approaches.

Response: This point was also caused by insufficient explanation. If one wishes to use the simulation-based algorithm instead, that is entirely possible. Moreover, the proposed method may even show its effect more clearly when combined with the simulation-based algorithm. We revised the explanation to emphasize this point.

In the experiments the paper explicitly mentions that FR is consistently outperformed by L-BFGS - then why we should use FR (even with the improved initial placement) at all? The paper does not make a good argument why we should choose that over the other, thus in a way making me question even the title of the paper.

Response: We believe that this concern also arises from the same ambiguity. The answer is that, regardless of the algorithm used to solve it, the FR force model itself is an important formulation with various applications. Again, we distinguish the simulation-based algorithm, which corresponds to the “FR algorithm” in the comment above, from the FR force model.

In general, it’s unclear what’s the impact of this research and how researchers could build upon it.

Response: Summarizing the discussion above, the impact of this paper and the ways in which other researchers can build upon it are as follows.

- Impact: For problems formulated using the FR force model, we propose an initial-placement method that can be effective regardless of whether a simulation-based algorithm or L-BFGS is used afterward, and we demonstrate that it is effective in various cases. Furthermore, in connection with Reviewer A’s comments, we also show that a similar idea can be applied to other graph drawing formulations.

- Building upon the work: The proposed method can be combined with other methods for solving problems based on the FR force model, such as multilevel approaches, or with other optimization methods. It is also relatively independent of subsequent studies on formulations for improving graph drawing quality.

We added explanations of these points, especially in the introduction.

B.4 Expansion of the Evaluation Dataset

The evaluation yields positive results, but in my opinion it could be further improved. First, I would suggest to expand 5.2 it in terms of number of graphs. More graphs from the Florida library could be used, or another source could be [1]. This expanded datasets already includes 3elt, but expands also on grids, cylinders, and other regular graphs that make up a solid benchmark set, and will provide a wider set of instances to discuss. If not all figures can be included in the main body of the paper, these could be moved in the supplemental material.

[1] Bartel, Gereon, et al. “An experimental evaluation of multilevel layout methods.” International Symposium on Graph Drawing. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010.

Response: Thank you for the suggestion. As suggested, we expanded the dataset in the Appendix to include additional graphs. We also substantially revised the presentation of the figures in response to this and other comments. Overall, we believe that the revised experiments demonstrate the effectiveness of the proposed method on a broader range of graphs.

We read the paper cited as [1] and examined the corresponding dataset. Unfortunately, the relevant links appear to be broken, and we were unable to obtain the dataset. We also checked the Internet Archive, but the download links were not functional, so we could not retrieve the full dataset. We also considered datasets used in papers cited by [1], and in fact experimented with some of them, including parts of the SNAP dataset (<https://snap.stanford.edu/data/index.html>). However, when the number of vertices exceeds ten thousand, the proposed method itself often still works without difficulty, but the subsequent iterations of the simulation-based algorithm or L-BFGS become very expensive and impractical. For such extremely large graphs, it seems necessary to combine the proposed method with more scalable methods such as multilevel approaches. We regard this as a promising direction for future work.

Although we ultimately decided not to include experiments on extremely large graphs and therefore cannot claim to provide a fully comprehensive benchmark set, we report that we expanded the dataset as much as possible to include a variety of graph types.

B.5 Visualization and Color Map

Concerning the visualization of the layout, I don’t believe that the index colouring adds much. A rainbow map, however, it’s not a great choice. If color should be kept, maybe I would consider using a sequential continuous color map with a legend (see [2] for examples).

[2] <https://matplotlib.org/stable/users/explain/colors/colormaps.html>

Response: Thank you for the suggestion. We agree that the current index-based coloring does not add much interpretive value. We revise the visualization by replacing the rainbow map with a sequential continuous color map, “viridis”.

B.6 Effect of Initial Placement Over Time

One thing that comes out of the charts in Figure 6 is that this different initial positioning has limited effect on the way the objective function decreases over time - which is an interesting phenomenon. In FR, the chart appears shifted below by how “better” the initial positioning is, with the non-CN version never catching up.

Response: We believe that this behavior is caused by the slow convergence of the FR algorithm, i.e., the simulation-based algorithm. For the graphs used in Figure 6, there appears to be a nearly unique equilibrium state, and the methods appear to converge toward the same best objective value for each graph. Of course, the initial placement and the subsequent algorithm affect the path through which the objective function decreases. In other words, if sufficiently many iterations are allowed, all methods seem to converge toward the same visually clean graph layout.

Therefore, we believe that it is somewhat inaccurate to say that the difference due to the initial placement remains permanently in the simulation-based algorithm. If a very large number of iterations is allowed, then even for the simulation-based algorithm, as for L-BFGS, the difference due to initialization gradually fades and the methods eventually converge to the same objective value. However, because the simulation-based algorithm converges slowly, a very large number of iterations would be required, and we stopped the plots at the current range because showing such a long run would make the figures difficult to read.

The discussion continues in the next response.

Differently, it seems that the objective functions of L-BFGS and CN-L-BFGS tend to converge before the iteration cap. Does this mean that the positive effect of the initial placement tends to fade faster with L-BFGS, despite the better initial condition?

Response: The same interpretation applies to this phenomenon. Independently of the merit of the proposed method itself, and simply as a comparison between existing methods, L-BFGS converges faster than the FR algorithm; that is, it approaches an optimal solution more rapidly. Once a method has essentially converged to an optimum, increasing the number of iterations has little effect on the graph layout, because the forces remain nearly balanced. If this optimal state is unique for a given graph, then there is naturally no difference in the final optimum caused by the initial placement. Thus, the effect of the initial placement is more closely related to how quickly the method approaches the optimal state. We revised the manuscript to explain these points clearly.

We also believe it is important to clarify that the objective-function values plotted in the chart do not perfectly correspond to visual quality. For example, the difference in objective value between a completely random state and a moderately clean layout can be much larger than the difference between a moderately clean layout and a fully clean layout. In continuous optimization, when the optimal objective value $\Phi(x^*)$ is known, it is common to plot the gap $\Phi(x) - \Phi(x^*)$ on a logarithmic scale. In our experiments, the optimal objective value is not known, so we simply plot the objective value itself. Therefore, even when the difference in objective value appears sufficiently small—that is, when the difference between RAND-LBFGS and CN-LBFGS appears small—there may still be a sufficiently visible difference in the graph layouts. We added an explanation of these points so that the discussion is more useful and convincing to readers who may have such questions.

B.7 Singularities in dwt_992 and collins_15NN

The dwt.992 and collins.15NN singularities could be described more. Would it be possible to generalize, highlighting what makes these two cases so different from each other?

Response: This point has not yet been addressed.

B.8 Spike in Objective Function at the Start of Optimization

Speaking of singularities, in the charts it seems like there is a spike in $f(X)$ right after the 0 seconds mark (like in 3elt, btree or cycle) is that an artifact?

Response: Thank you for pointing this out. This phenomenon occurs because, in the initial stage of the FR algorithm, the step size is too large and the solution oscillates. Our original intention was to conduct experiments using exactly the same parameters as external implementations such as NetworkX in order to make the experiments as natural as possible. In that sense, this was intended behavior.

However, we agree that it can easily cause misunderstanding or questions. If properly explained, adjusting the initial step size is also a natural choice for conducting more stable numerical experiments. Therefore, in the revised manuscript, we adjusted the initial step size of the FR algorithm and substantially reduced the oscillation. As a result, we believe that the effect of the proposed method is now easier to see.

B.9 Organization of Sections 6 and 7

I am less persuaded from Sections 6 and 7. Section 6 includes considerations that I would include in Section 4, as to further justify your optimization design. I would also simplify the discussion, removing subsections and integrating their contents in Section 4.

Response: Thank you for the suggestion. We reconsidered the section organization seriously and revised the manuscript to make the structure clearer. First, we agreed that the material on the KK layout and related topics from Section 6 was more appropriate for Section 4, so we created a new Section 4.5 and moved that discussion there. We believe that this makes it clearer which aspects of the problem the proposed method focuses on. However, moving all of Section 6 into Section 4 would have pushed the experimental results and other material that we wanted to emphasize further toward the end of the paper. For this reason, after consideration, we kept part of Section 6 in place. We also simplified Section 4 by removing some redundant material. Although the revision may not follow the suggestion completely, we believe that it improves the overall quality of the manuscript.

In Section 7 I would focus on the impact of this approach on graph drawing more specifically. For example, where is the next step for this technique, in the context of graph drawing? Will its presence in a drawing pipeline make a difference, or the gains in quality are not sufficient to justify its use? What about its inclusion in a multi-level pipeline (see also a similar point above)?

Response: In connection with Reviewer A's comments, we removed part of Section 7 and moved part of it to Section 4. The previous Section 7.1 gave only a broad suggestion of possible applications. We revised the manuscript so that generalizations to layouts such as HC and Eades are

now explained clearly, especially in Section 4.5. We believe that this revision shows more explicitly that the proposed method is also effective for such graph drawing problems, and also explains what properties a problem should have for the current method to be effective. Since we also regard multilevel approaches as one of the most promising applications of the proposed method, we added a new Section 6.1 to discuss this point.

B.10 Moving the Section 8 to the Appendix

I would move Section 8 to supplemental material out of the main body of the paper.

Response: Thank you. In the revision, we moved the implementation-oriented material (NetworkX implementation details) to the Appendix so that the main narrative focuses on the method, experiments, and discussion.

B.11 Bibliography

Bibliography must be updated. Some papers do not include any venue, others do not have a DOI, others only report their archive version (while journal versions are available).

Response: Thank you for the careful note. In the revision, we reviewed the bibliography entries and improved completeness where possible (e.g., adding missing metadata and cleaning local-only fields), and we checked for cases where a journal/conference version should be cited instead of an archive-only version.

Closing

We again thank the editor and the reviewers for their valuable comments. We believe that the revised manuscript is clearer in scope, better positioned with respect to existing force-directed graph drawing methods, and more explicit about both the strengths and limitations of the proposed approach.

Sincerely,

Hiroki Hamaguchi, Naoki Marumo, and Akiko Takeda